Evaluating normality + normal probability plot

1 RECOGNISE IT

- Asked whether a normal model/t method is plausible.
- Raw data, histogram, boxplot or normal probability plot is supplied.
- Need an evidence-based normality judgement.

2 DECISION / METHOD

1. Check shape/symmetry and outliers.
2. Compare mean and median; skewness/kurtosis near 0 supports normality.
3. Build/read normal probability plot.
4. Approximately straight positive line -> approximately normal.

3 FORMULA

```
ordered x_(j)
alpha_j = j/(n+1)
z_j = Φ^(-1)(alpha_j)
plot x-axis z_j, y-axis ordered x_(j)
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: nothing for normality plotting.
Still know: straight-line criterion, axes, probability positions, and that one check alone is not proof.

5 EXCEL ROUTE

- Sort X ascending.
- `n=COUNT(range)`; j=1,...,n.
- `α=j/(n+1)`.
- `=NORM.S.INV(alpha)`.
- Insert XY scatter: x=Z, y=ordered X.

6 TINY WORKED EXAMPLE

With n=3 and ordered X={4,5,7}, α values are .25,.50,.75 and Z values about -0.674,0,0.674. Plot (Z,X) and inspect linearity.

7 TOP EXAM TRAPS

- Using NORM.S.DIST(α,TRUE) instead of inverse.
- Swapping axes.
- Calling any upward trend normal despite strong curvature/outliers.

8 SAY ALOUD

Normal data make an approximately straight, positive-slope normal probability plot.

9 EXIT CHECK

Which Excel function converts alpha_j into the theoretical z_j?

ONE-GLANCE VISUAL ROUTE

